

Lecture/Week 6

Interpolation

Numerical Method to approximate numbers or functions over a given interval $[a, b]$

Newton's Divided Difference Interpolation MethodGiven Data

$$(x_0, f_0) (x_1, f_1) (x_2, f_2) \dots (x_n, f_n)$$

$n+1$ data points

- Approximate a function from a given set of data points
- function will be a polynomial of degree ' n '. that is,

$$f(x) = P_n(x)$$

Formula

$$f(x) = P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) \dots$$

$$\dots a_n(x-x_0)(x-x_1)(x-x_2) \dots (x-x_{n-1}) \quad \text{--- (1)}$$

In (1) a_i 's for $i = 0, 1, 2, \dots, n$

are ~~are~~ unknown. ~~for~~ These are called Newton's divided difference coefficients.

(2)

Task is to find a_i 's and substitute them in (1) to get the polynomial.

We know Interpolation Conditions

$$P_n(x_i) = f_i \quad i=0, 1, 2, \dots, n.$$

Eg. $P_n(x_0) = f_0$; $P_n(x_1) = f_1$; $P_n(x_2) = f_2$; $\dots$ $P_n(x_n) = f_n$

Using $P_n(x_0) = f_0$ in (1)

$$P_n(x_0) = a_0 + a_1(x_0 - x_0) + a_2(x_0 - x_0)(x_0 - x_1) + a_3(x_0 - x_0)(x_0 - x_1)(x_0 - x_2) + \dots + a_n(x_0 - x_0)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_{n-1}) = f_0$$

$$\boxed{a_0 = f_0}$$

Similarly $P_n(x_1) = f_1$ in (1)

$$P_n(x_1) = a_0 + a_1(x_1 - x_0) + a_2(x_1 - x_0)(x_1 - x_1) + a_3(x_1 - x_0)(x_1 - x_1)(x_1 - x_2) + \dots + a_n(x_1 - x_0)(x_1 - x_1)(x_1 - x_2) \dots (x_1 - x_{n-1}) = f_1$$

$$a_0 + a_1(x_1 - x_0) = f_1$$

$$\boxed{a_1 = \frac{f_1 - f_0}{x_1 - x_0}}$$

where $a_0 = f_0$

Now $P_n(x_2) = f_2$

$$P_n(x_2) = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1) + a_3(x_2 - x_0)(x_2 - x_1)(x_2 - x_2) + \dots + a_n(x_2 - x_0)(x_2 - x_1)(x_2 - x_2) \dots (x_2 - x_{n-1}) = f_2$$

(3)

$$a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1) = f_2$$

$$a_2 = \frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0}$$

Similarly one can find $a_3, a_4, \dots, a_n$.

However process is difficult and computationally expensive. Therefore we use the difference brackets to find the a_i 's.

Given

(4)

$$(x_0, f_0) (x_1, f_1) (x_2, f_2) \dots (x_n, f_n)$$

$$f(x) = P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) + \dots + a_n(x-x_0)(x-x_1)(x-x_2)\dots(x-x_{n-1}) \quad \text{--- (1)}$$

where

$$a_0 = f_0$$

$$a_1 = f[x_1, x_0]$$

$$a_2 = f[x_2, x_1, x_0]$$

$$a_3 = f[x_3, x_2, x_1, x_0]$$

⋮

$$a_{n-1} = f[x_{n-1}, x_{n-2}, \dots, x_0]$$

$$a_n = f[x_n, x_{n-1}, x_{n-2}, \dots, x_0]$$

General formula

$$a_m = f[x_m, x_{m-1}, \dots, x_0]$$

$$= \frac{f[x_m, x_{m-1}, \dots, x_0] - f[x_{m-1}, x_{m-2}, \dots, x_0]}{x_m - x_0}$$

Example

(5)

(1, 1.54) (2, 0.58) (3, 0.01) (4, 0.35)

Construct divided difference polynomial to approximate $x = 2.6$.

x	f	$f[x_i, x_0]$	$f[x_2, x_1, x_0]$
$x_0 \leftarrow 1$	$f_0 \leftarrow 1.54$	$\frac{f_1 - f_0}{x_1 - x_0} = \frac{0.58 - 1.54}{2 - 1} = -0.96$	$\frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0} = \frac{-0.96 - (-0.57)}{3 - 1} = -0.195$
$x_1 \leftarrow 2$	$f_1 \leftarrow 0.58$	$\frac{f_2 - f_1}{x_2 - x_1} = \frac{0.01 - 0.58}{3 - 2} = -0.57$	$\frac{0.29 - (-0.57)}{4 - 2} = 0.43$
$x_2 \leftarrow 3$	$f_2 \leftarrow 0.01$	$\frac{f_3 - f_2}{x_3 - x_2} = \frac{0.3 - 0.01}{4 - 3} = 0.29$	
$x_3 \leftarrow 4$	$f_3 \leftarrow 0.3$		

$$f[x_3, x_2, x_1, x_0]$$

$$\frac{0.43 - 0.2}{4 - 1} = 0.0767$$

$P_3(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$
using values from table and given data.

$$P_3(x) = 1.54 - 0.96(x-1) + 0.2(x-1)(x-2) + 0.0767(x-1)(x-2)(x-3)$$

Ans.

$$P_3(2.6) = f(2.6) = 1.54 - 0.96(2.6-1) + 0.2(2.6-1)(2.6-2) + 0.0767(2.6-1)(2.6-2)(2.6-3)$$

$$= \boxed{\quad} \text{ Ans.}$$

(6)

Example (2)

The upward velocity of a rocket is given as a function of time

$t(\text{sec})$	0	10	15	20	22.5	30
$V(\text{m/sec})$	0	227.04	362.78	517.35	602.97	901.67

- Determine the velocity at $t=16$ sec using third order polynomial Interpolation.
- Find the distance covered by rocket from $t=11$ to $t=16$ sec.
- Find acceleration of the rocket at $t=16$ sec.

Solution

t	V
0	0
10	227.04
15	362.78
$\rightarrow t=16$	$\rightarrow$
20	517.35
22.5	602.97
30	901.67

t	V
$t_0 \leftarrow 10$	227.04 $\rightarrow v_0$
$t_1 \leftarrow 15$	362.78 $\rightarrow v_1$
$t_2 \leftarrow 20$	517.35 $\rightarrow v_2$
$t_3 \leftarrow 22.5$	602.97 $\rightarrow v_3$

(7)

t	V	a_0	$V[t_1, t_0]$	a_1	$V[t_2, t_1, t_0]$	a_2	$V[t_3, t_2, t_1, t_0]$
10	227.04			27.148			
15	362.78		$\frac{362.78 - 227.04}{15 - 10} = 27.148$		$\frac{30.914 - 27.148}{20 - 10} = 0.37660$		
20	517.35		$\frac{517.35 - 362.78}{20 - 15} = 30.914$		$\frac{34.248 - 30.914}{22.5 - 20} = 0.4453$		$\frac{0.44453 - 0.37660}{22.5 - 10} = 5.4347 \times 10^{-3}$
22.5	602.97		$\frac{602.97 - 517.35}{22.5 - 20} = 34.248$				

$$V = f(x) = P_3(x) = a_0 + a_1(t - t_0) + a_2(t - t_0)(t - t_1) + a_3(t - t_0)(t - t_1)(t - t_2)$$

$$V(t) = 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15) + 5.4347 \times 10^{-3}(t - 10)(t - 15)(t - 20)$$

Ans.

a) $V(16) = 227.04 + 27.148(16 - 10) + 0.37660(16 - 10)(16 - 15) + 5.4347 \times 10^{-3}(16 - 10)(16 - 15)(16 - 20) = \underline{\underline{392.06 \text{ m/sec}}}$

b) $S(16) - S(11) = \int_{11}^{16} V(t) dt$

$$= \int_{11}^{16} 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15) + 5.4347 \times 10^{-3}(t - 10)(t - 15)(t - 20) dt$$

$$= 1605 \text{ m}$$

c) $a(16) = \frac{dV}{dt} \bigg|_{t=16}$

$$= 29.664 \text{ m/sec}^2$$

(8)

Newton's Forward Difference Interpolation

- Data points are equally spaced

$$x_n - x_{n-1} = h$$

Eg.

x	f	t	v
2	6	0	10
4	5	3	15
6	-8	6	20
8	1	9	60

$$P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$$

$$= f_0 + \frac{f_1 - f_0}{x_1 - x_0} (x - x_0) + \frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0} (x - x_0)(x - x_1)$$

$$= f_0 + \frac{f_1 - f_0}{h} (x - x_0) + \frac{(\frac{f_2 - f_1}{2h} - \frac{f_1 - f_0}{h})}{2h^2} (x - x_0)(x - x_1) + \dots$$

General formula.

~~$$P_n(x) = f_0 + \frac{\Delta f_0}{h} + \frac{\Delta^2 f_0}{2! h^2}$$~~

General formula

$$P_n(x) = f_0 + \frac{\Delta f_0}{1! h} (x-x_0) + \frac{\Delta^2 f_0}{2! h^2} (x-x_0)(x-x_1) + \frac{\Delta^3 f_0}{3! h^3} (x-x_0)(x-x_1)(x-x_2) + \frac{\Delta^4 f_0}{4! h^4} (x-x_0)(x-x_1)(x-x_2)(x-x_3) + \dots$$

Example

x	f	Δf	$\Delta^2 f$	$\Delta^3 f$	$\Delta^4 f$	$\Delta^5 f$
2	-7					
4	-3	-3 - (-7) = 4				
6	6	6 - (-3) = 9	9 - 4 = 5			
8	25	25 - 6 = 19	19 - 9 = 10	10 - 5 = 5		
10	62	62 - 25 = 37	37 - 19 = 18	18 - 10 = 8	8 - 5 = 3	
12	129	129 - 62 = 67	67 - 37 = 30	30 - 18 = 12	12 - 8 = 4	4 - 3 = 1

h = 2

$$P_5(x) = f_0 + \frac{\Delta f_0}{2} (x-x_0) + \frac{\Delta^2 f_0}{2! (2)^2} (x-x_0)(x-x_1) + \frac{\Delta^3 f_0}{3! (2)^3} (x-x_0)(x-x_1)(x-x_2) + \frac{\Delta^4 f_0}{4! (2)^4} (x-x_0)(x-x_1)(x-x_2)(x-x_3) + \frac{\Delta^5 f_0}{5! (2)^5} (x-x_0)(x-x_1)(x-x_2)(x-x_3)(x-x_4)$$

$$= -7 + \frac{4}{2} (x-2) + \frac{5}{16} (x-2)(x-4) + \frac{5}{3! (2)^3} (x-2)(x-4)(x-6) + \frac{3}{4! (2)^4} (x-2)(x-4)(x-6)(x-8) + \frac{1}{5! (2)^5} (x-2)(x-4)(x-6)(x-8)(x-10)$$

Ans.